Passivity-based Control of a Double-Pendulum Crane using Propagation of Vibration Suppression

Ho Duc Tho[a,*], Kazuhiko Terashima[b], Takanori Miyoshi[c]

[a] *Department of Control Engineering and Automation, Ho Chi Minh City University of Technology (HCMUT), 268 Ly Thuong Kiet Street, District 10, Ho Chi Minh City, Vietnam*
[b] *In memory of Prof. Kazuhiko Terashima, who passed away in May 2024*
[c] *Nagaoka University of Technology, 1603-1 Kamitomiokamachi, Nagaoka, Niigata 940-2137, Japan*

Abstract

The second swing angle information may not be strictly required to fully control the double-pendulum overhead crane (DPOC) system. Such an insight stems from the propagation of vibration suppression principle, which states that if either of the two swing angles is suppressed by control, both pendulums will cease to vibrate altogether. This intuition serves as the basis of the augmented passivity-based control law. The trolley velocity—the default command of most industrial cranes—is chosen as the system's control input. Passivity properties of the system and the control law design are formulated based on a linearized model. The region of attraction (ROA) of the nonlinear closed-loop system is then computed using the sum-of-square (SOS) programming. The established controller does not require measurements or estimations of any rate or velocity signals. In the absence of the second swing angle information, the proposed approach can achieve similar control performance compared to existing full-state feedback controllers. An experimental result on a small-scale DPOC rig is also provided.

Keywords: Crane, underactuated system, passivity, vibration control, vibration propagation.

1. Introduction

Double-pendulum phenomenon occurs in crane systems when the discrepancy between the natural frequencies of the payload's first and second vibration modes is significant (Arabasi and Masoud, 2022; Vaughan et al., 2010; Masoud and Alhazza, 2013). Such an effect happens if the hook's mass is non-negligible relative to the payload's mass and/or when the payload has a sizable dimension in length. In comparison to single-pendulum cranes (Sun et al., 2013; Qian et al., 2017; Bartolini et al., 2002), the underactuation degree of DPOCs increases by one, adding extra complexity to the control law design. The transportation process using overhead cranes with the double-pendulum configuration is commonly found in practice, and hence, is practically relevant. Moreover, feedback control designs for DPOCs are also theoretically challenging due to their intricate dynamics with a high level of underactuation. For these reasons, the vibration suppression control of DPOCs using feedback controllers has been extensively investigated in recent years. The DPOC provides a fertile ground for applying various control techniques on a practically meaningful system: energy and passivity-based control (Sun et al., 2019; Qian, 2018); sliding-mode control (Tuan and Lee, 2013; Ouyang et al., 2019); adaptive control (Zhang et al., 2016; Lu et al., 2019; Yang et al., 2023); model predictive control (Thomas and Sawodny, 2022); nonlinear optimal control (Rigatos, 2023; Li et al., 2023); back-stepping control (Miao et al., 2023).

In fact, the DPOC belongs to a more general class of mechanically underactuated systems (Ortega et al., 2010; Fantoni and Lozano, 2002; Lu and Fang, 2021; Gnucci and Marino, 2021; Romero et al., 2018; Donaire et al., 2016). Therefore, newly discovered properties, phenomena, and control designs/challenges for the DPOC may inductively be generalized to a much wider class of underactuated systems. In this paper, by using the DPOC as a vehicle, it will be shown that information of all unactuated coordinates might not absolutely be required for control, i.e., some of them might be disregarded in the control law without jeopardizing the system stability and control performance.

To fully regulate the DPOC system, the existing studies always mandate three pieces of information: the trolley position, the first swing angle, and the second swing angle. As our main contribution, we will demonstrate that direct measurement (via a sensor) or indirect estimation (by a state observer) of the second swing angle is not absolutely needed, i.e., only the trolley position and the first swing angle information are required for control. Such an insight is based on the propagation of vibration suppression principle (see section 3), which partly states that if the first swing angle vanishes, so does the second swing

*Corresponding author. Tel. +84-0901-352-291. This paper was not presented at any IFAC meeting.
Email addresses: hoductho@hcmut.edu.vn (Ho Duc Tho), miyoshi@mech.nagaokaut.ac.jp (Takanori Miyoshi)

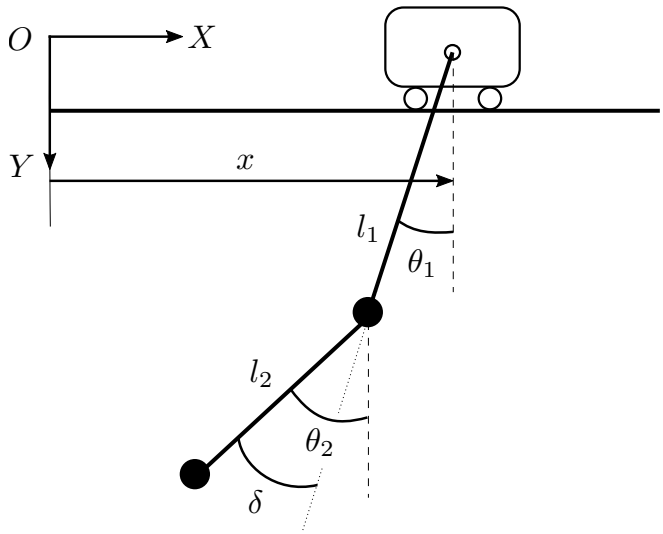

Figure 1: A mathematical model of a DPOC.

angle. Consequently, it is safe to exclude the second swing angle's measurement from the control law. By eliminating the need of the second swing angle information, we can bypass the use of an actual sensor (helping reduce the cost, maintenance effort, and device failure risk) or a state observer (which helps avoid inaccurate state estimation caused by input noises, unmodeled dynamics, disturbances, and parametric uncertainties). Therefore, the cost effectiveness and the system resiliency can be enhanced as a whole. As opposed to the existing studies that employed force control input, the commanding signal of the DPOC resulted from the proposed controller is the trolley's velocity, which is the default control mode of most industrial cranes in use. The proposed control law also does not require any velocity or rate signals (see section 4). In comparison with prevailing full-state feedback controllers (including the passivity-based control (Sun et al., 2019), the sliding-mode control (Tuan and Lee, 2013), and the adaptive fuzzy control (Yang et al., 2022)), control peformance of the established controller does not deteriorate in the absence of the second swing angle's measurement, confirming the primary contribution of the paper.

Technically, based on a linearized model, passivity properties of the DPOC system are identified (see section 4.1). The proposed control law is then established, and the local stability analysis is conducted by employing the Lyapunov linearization theorem (see section 4.2). Moreover, to quantify the stability result, a ROA computation for the original nonlinear closed-loop system is performed using the SOS programming (see section 4.3).

Notations. $\mathbb{N}$, $\mathbb{N}^+$ and $\mathbb{R}^n$ denote the set of non-negative integers, positive integers, and the n-th dimensional Euclidean space, respectively. The complex and time variables are represented by s and t, respectively. The set $\mathcal{L}_\infty$ contains all uniformly bounded functions on $[0,\infty)$. Moreover, $\mathcal{L}_\infty^{\to 0}$ denotes the set of all functions that both belong to $\mathcal{L}_\infty$ and are asymptotically decreasing to zero as $t \to \infty$. The set of all time-weighted (TW) exponentially decaying functions of the form $\sum_{i=1}^{r}\sum_{k=1}^{m_i} t^{k-1} R_{ik} \exp(-\gamma_{ik} t)\sin(\omega_{ik} t + \varphi_{ik})$ (for some $r, m_i \in \mathbb{N}^+$, $\gamma_{ik} > 0$, and $R_{ik}, \omega_{ik}, \varphi_{ik} \in \mathbb{R}$) is denoted as $\mathcal{E}$. The n-th order derivative ($n \in \mathbb{N}$) of a function $h(t)$ with respect to the time variable t is represented by $h^{(n)}(t)$. Furthermore, the $m\times n$ real, $m\times n$ zero, and $m\times m$ square identity matrices are denoted by $\mathbb{R}^{m\times n}$, $\mathbf{0}_{m\times n}$ and $\boldsymbol{I}_{m\times m}$, respectively.

2. Problem formulation

In this section, a mathematical model of the DPOC is firstly presented. Control objectives and requirements are then stated.

An abstracted DPOC system is depicted in Fig. 1. The trolley position, the first swing angle, and the second swing angle are denoted as $x(t)$ (m), $\theta_1(t)$ (rad), and $\theta_2(t)$ (rad), respectively. The system parameters consist of m_1 (weight of the hook, in kg), m_2 (weight of the payload, in kg), l_1 (length of the first pendulum, in m), l_2 (length of the second pendulum, in m), and g (gravitational acceleration, $g = 9.81$ m/s^2). The Euler-Lagrange equation applied on the unactuated coordinates θ_1 and θ_2 results in the following nonlinear dynamics (Sun et al., 2019; Tuan and Lee, 2013)

$$\begin{aligned}
&(m_1+m_2)\, l_1\ddot{\theta}_1 + m_2 l_2 \cos(\theta_1-\theta_2)\,\ddot{\theta}_2 \\
&\quad + m_2 l_2 \sin(\theta_1-\theta_2)\,\dot{\theta}_2^2 + (m_1+m_2)\, g \sin\theta_1 \\
&\quad = (m_1+m_2)\cos\theta_1 \ddot{x} \qquad (1)\\
&l_1\cos(\theta_1-\theta_2)\,\ddot{\theta}_1 + l_2\ddot{\theta}_2 - l_1\sin(\theta_1-\theta_2)\,\dot{\theta}_1^2 \\
&\quad + g\sin\theta_2 = \cos\theta_2\ddot{x}.
\end{aligned}$$

The linearized model of the nonlinear DPOC system (1) around its usual equilibrium point is given by

$$\begin{aligned}
&(m_1+m_2)\, l_1\ddot{\theta}_1 + m_2 l_2\ddot{\theta}_2 + (m_1+m_2)\, g\theta_1 = (m_1+m_2)\,\ddot{x} \\
&l_1\ddot{\theta}_1 + l_2\ddot{\theta}_2 + g\theta_2 = \ddot{x}. \qquad (2)
\end{aligned}$$

For DPOC systems, it is desirable to achieve a precise transportation process without residue vibration to enhance productivity and safety. Therefore, the following control objectives should be fulfilled:

1) The trolley asymptotically reaches its constant reference $x_d \in \mathbb{R}$ at zero terminal speed, i.e.,

$$\lim_{t\to\infty} x(t) = x_d,\ \lim_{t\to\infty} \dot{x}(t) = 0. \qquad (3)$$

2) Both swing angles cease, that is to say

$$\lim_{t\to\infty} \theta_i(t) = \lim_{t\to\infty} \dot{\theta}_i(t) = 0\ (i = 1, 2). \qquad (4)$$

In addition to (3)–(4), further requirements for the to-be-designed controller are:

a) The control law should not contain the second swing angle information.

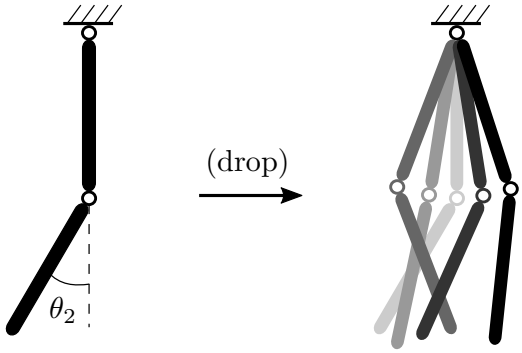

Figure 2: Vibration propagation through pendulums.

b) The control law should not use any derivative signals.
c) The control input should be the trolley's velocity.

In order to address all of the aforementioned control objectives and requirements, we firstly introduce the propagation of vibration suppression principle (see section 3), which states that if one of the two swing angles vanishes, so does the other. Such a principle allows us to safely disregard the second swing angle information. It also serves as a premise of the augmented passivity-based control law (see section 4).

3. Propagation of vibration suppression

We firstly demonstrate that the *vibration (excitation)* can be propagated from bodies to bodies. This motivates us to speculate that the *vibration suppression* can also be propagated from bodies to bodies. The propagation of vibration suppression principle will be precisely described in Theorems 1–2.

In the setup shown in Fig. 2, the first pendulum is at its equilibrium state $\theta_1(0) = \dot{\theta}_1(0) = 0$, i.e., it is not exhibiting any vibration. The second pendulum is then dropped at zero speed from a non-zero beginning position, imposing an initial vibration on it. Under such a circumstance, it can be perceived that the overall system will start to oscillate indefinitely (assuming there is no friction). On the other hand, if the second pendulum is dropped at zero speed from a zero beginning position (i.e., no initial vibration is forced), the overall system will continue staying at rest. Therefore, the physical intuition suggests that the initial vibration of the second pendulum propagates through the first pendulum.

Since the vibration excitation can be propagated, it is reasonable to speculate that the inverse process—the vibration suppression—can be propagated as well. Indeed, this speculation is true, and it will subsequently be established for both nonlinear DPOC model in Theorem 1 and linearized DPOC system in Theorem 2.

Theorem 1 (Propagation of vibration suppression for the nonlinear model)**.** *Consider the nonlinear DPOC system (1). Assume that, for all $t \geq 0$, $\theta_i^{(j)}(t)$ is continuous and the swing angles $\theta_i(t) \in (-\pi/2,\ \pi/2)$ ($i = 1, 2$ and $j = 0, 1, 2$). If $\ddot{x} \in \mathcal{L}_\infty^{\to 0}$, then the following result holds*

$$\theta_1^{(j)}(t) \in \mathcal{L}_\infty^{\to 0} \iff \theta_2^{(j)}(t) \in \mathcal{L}_\infty^{\to 0},\ (j = 0, 1, 2).$$

Proof. Due to lack of space, the proof is only carried out for the forward direction ($\Rightarrow$). The reverse direction can be proven in the same manner. Assuming that $\theta_1^{(j)}(t) \in \mathcal{L}_\infty^{\to 0}$ ($j = 0, 1, 2$). Substituting this assumption as well as the hypothesis $\ddot{x}(t) \to 0$ as $t \to \infty$ into (1), it is true that

$$\lim_{t\to\infty} \left[\cos(\theta_1 - \theta_2)\,\ddot{\theta}_2 + \sin(\theta_1 - \theta_2)\,\dot{\theta}_2^2\right] = 0 \tag{5}$$

$$\lim_{t\to\infty} \left(l_2\ddot{\theta}_2 + g\sin\theta_2\right) = 0, \tag{6}$$

where the Squeeze theorem (Sherbert and Bartle, 2011, p. 114) has been used together with the fact that $|\sin(.)| \leq 1$ and $|\cos(.)| \leq 1$, $\forall(.) \in \mathbb{R}$. From (6), the swing jerk $\ddot{\theta}_2$ cannot blow up to infinity as $t \to \infty$ because $|\sin\theta_2| \leq 1$, and thus, $\ddot{\theta}_2 \in \mathcal{L}_\infty$ due to the assumption that $\dot{\theta}_2$ is continuous. From the first equation of (1), the result $\dot{\theta}_2 \in \mathcal{L}_\infty$ holds since it is assumed that $\ddot{\theta}_1, \ddot{x} \in \mathcal{L}_\infty$. Because $\dot{\theta}_2, \ddot{\theta}_2 \in \mathcal{L}_\infty$, there exist $K_1 > 0$ and $K_2 > 0$ such that $|\dot{\theta}_2(t)| \leq K_1$ and $|\ddot{\theta}_2(t)| \leq K_2$, $\forall t \geq 0$. The following term is evaluated next

$$\begin{aligned}
&\left|(\cos\theta_2 - \cos(\theta_1 - \theta_2))\,\ddot{\theta}_2 + (\sin(\theta_2 - \theta_1) - \sin\theta_2)\,\dot{\theta}_2^2\right| \\
&\leq |\cos\theta_2 - \cos(\theta_1 - \theta_2)|\,|\ddot{\theta}_2| + |\sin(\theta_2 - \theta_1) - \sin\theta_2|\,\dot{\theta}_2^2 \\
&\leq 2\left|\sin\frac{\theta_1}{2}\right|\left(K_1 + K_2^2\right) \leq |\theta_1|\left(K_1 + K_2^2\right),
\end{aligned} \tag{7}$$

where the following results have been used: $\cos\theta_2 - \cos(\theta_1 - \theta_2) = -2\sin(\theta_1/2)\sin(\theta_2 - \theta_1/2)$, $\sin(\theta_2 - \theta_1) - \sin\theta_2 = -2\sin(\theta_1/2)\cos(\theta_2 - \theta_1/2)$, $|\cos(.)| \leq 1$, $|\sin(.)| \leq 1$, and $|\sin(.)| \leq |(.)|$, $\forall(.) \in \mathbb{R}$. Let $\varepsilon > 0$. Since $\lim_{t\to\infty}\theta_1(t) = 0$, there exists $T(\varepsilon) > 0$ such that for all $t > T(\varepsilon)$, we have $|\theta_1(t)| \leq \varepsilon/\left(K_1 + K_2^2\right)$. Thus, from (7)

$$\begin{aligned}
&\left|(\cos\theta_2 - \cos(\theta_1 - \theta_2))\,\ddot{\theta}_2 + (\sin(\theta_2 - \theta_1) - \sin\theta_2)\,\dot{\theta}_2^2\right| \\
&\leq \left(\varepsilon/\left(K_1 + K_2^2\right)\right)\left(K_1 + K_2^2\right) = \varepsilon.
\end{aligned}$$

As a result, by definition (Sherbert and Bartle, 2011, p. 121)

$$\begin{aligned}
\lim_{t\to\infty} \Big[&(\cos\theta_2 - \cos(\theta_1 - \theta_2))\,\ddot{\theta}_2 \\
&+ (\sin(\theta_2 - \theta_1) - \sin\theta_2)\,\dot{\theta}_2^2\Big] = 0.
\end{aligned} \tag{8}$$

The term $\cos\theta_2\ddot{\theta}_2 - \sin\theta_2\dot{\theta}_2^2$ can be evaluated as follows

$$\begin{aligned}
&\cos\theta_2\ddot{\theta}_2 - \sin\theta_2\dot{\theta}_2^2 = \\
&\left[(\cos\theta_2 - \cos(\theta_1 - \theta_2))\,\ddot{\theta}_2 + (\sin(\theta_2 - \theta_1) - \sin\theta_2)\,\dot{\theta}_2^2\right] \\
&+ \left[\cos(\theta_1 - \theta_2)\,\ddot{\theta}_2 + \sin(\theta_1 - \theta_2)\,\dot{\theta}_2^2\right].
\end{aligned} \tag{9}$$

Substituting (5) and (8) into (9) yields

$$\lim_{t\to\infty} l_2 \left(\cos\theta_2\ddot{\theta}_2 - \sin\theta_2\dot{\theta}_2^2\right) = 0. \tag{10}$$

Because $|\cos\theta_2(l_2\ddot{\theta}_2 + g\sin\theta_2)| \leq |l_2\ddot{\theta}_2 + g\sin\theta_2|$ (as $|\cos\theta_2| \leq 1$) and $\lim_{t\to\infty}(l_2\ddot{\theta}_2 + g\sin\theta_2) = 0$ (see (6)), by the Squeeze theorem, it is true that

$$\lim_{t\to\infty} \left(l_2\cos\theta_2\ddot{\theta}_2 + g\sin\theta_2\cos\theta_2\right) = 0. \tag{11}$$

Subtracting (10) by (11) results in

$$\lim_{t\to\infty} \left[\sin\theta_2 \left(g\cos\theta_2 + l_2\dot{\theta}_2^2\right)\right] = 0. \tag{12}$$

Since it is assumed that $\theta_2 \in (-\pi/2,\ \pi/2)$, there is $K_3 > 0$ such that $0 < K_3 < \cos\theta_2 \leq 1$, and hence

$$|\sin\theta_2| = \frac{1}{gK_3}|gK_3\sin\theta_2| \leq \frac{1}{gK_3}\left|\sin\theta_2\left(g\cos\theta_2 + l_2\dot{\theta}_2^2\right)\right|,$$

which implies that $\lim_{t\to\infty}\sin\theta_2 = 0$ due to (12) and the Squeeze theorem. Because $\theta_2(t) \in (-\pi/2,\ \pi/2)$ is continuous by assumption, the following operation can be performed: $\lim_{t\to\infty}\sin\theta_2(t) = 0 \Rightarrow \sin\left(\lim_{t\to\infty}\theta_2(t)\right) = 0 \Rightarrow \lim_{t\to\infty}\theta_2(t) = 0$. Thus, $\theta_2 \in \mathcal{L}_\infty^{\to 0}$. Furthermore, since $\lim_{t\to\infty}\sin\theta_2 = 0$, the result $\lim_{t\to\infty}\ddot{\theta}_2 = 0$ follows from (6), and hence $\ddot{\theta}_2 \in \mathcal{L}_\infty^{\to 0}$. Since it was proven that $\lim_{t\to\infty}\theta_2 = 0$ and $\ddot{\theta}_2 \in \mathcal{L}_\infty$, it is true that $\lim_{t\to\infty}\dot{\theta}_2 = 0$ due to the Barbalat Lemma (Khalil, 2002, p. 323). As a result, $\dot{\theta}_2 \in \mathcal{L}_\infty^{\to 0}$. This therefore completes the proof. □

The case of linearized model will be considered next, in which both asymptotically decaying and TW exponentially decaying scenarios will be formulated.

Theorem 2 (Propagation of vibration suppression for the linearized model)**.** *Consider the linearized DPOC system (2), the following statements hold true*

1) *(Asympt. decaying case) Assume that $\ddot{x}(t) \in \mathcal{L}_\infty^{\to 0}$, then*

$$\theta_1^{(j)}(t) \in \mathcal{L}_\infty^{\to 0} \iff \theta_2^{(j)}(t) \in \mathcal{L}_\infty^{\to 0},\ (j = 0, 1, 2).$$

2) *(TW exp. decaying case) If $\theta_1(t), \ddot{x}(t) \in \mathcal{E}$ or $\theta_2(t) \in \mathcal{E}$, then $\theta_1^{(n)}(t) \in \mathcal{E}$ and $\theta_2^{(n)}(t) \in \mathcal{E}$ for all $n \in \mathbb{N}$.*

Proof. The proof for statement 1) will be carried out for the forward direction ($\Rightarrow$). The reverse direction can be proven in the same manner. By substituting the assumption $\ddot{x}(t), \theta_1^{(j)}(t) \in \mathcal{L}_\infty^{\to 0}$ ($j = 0, 1, 2$) into (2), it can be obtained that $\theta_2(t) \in \mathcal{L}_\infty^{\to 0}$ and $\ddot{\theta}_2(t) \in \mathcal{L}_\infty^{\to 0}$. This further implies that $\dot{\theta}_2(t) \to 0$ as $t \to \infty$ due to the Barbalat lemma (Farkas and Wegner, 2016). In addition, since $\ddot{\theta}_2$ is uniformly bounded on $[0, \infty)$, it is true that $\dot{\theta}_2$ is Lipschitz continuous on the same interval (Sherbert and Bartle, 2011, p. 179); thus, $\dot{\theta}_2$ belongs to $\mathcal{L}_\infty^{\to 0}$. The proof for item a) is therefore completed.

The statement 2) will be proved next. From (2), the following results hold

$$\theta_1 = \frac{m_1 l_2\ddot{\theta}_2 + (m_1 + m_2)\, g\theta_2}{(m_1 + m_2)\, g} \tag{13}$$

$$\theta_2 = \frac{m_1 l_1\ddot{\theta}_1 + (m_1 + m_2)\, g\theta_1 - m_1\ddot{x}}{m_2 g}. \tag{14}$$

If it is assumed that $\theta_1(t), \ddot{x}(t) \in \mathcal{E}$, we have $\theta_1^{(n)}(t) \in \mathcal{E}$ $\forall n \in \mathbb{N}$. It is then inferred from (14) that $\theta_2(t) \in \mathcal{E}$, and thus, $\theta_2^{(n)}(t) \in \mathcal{E}$ $\forall n \in \mathbb{N}$. This completes the proof for the first part of statement 2). For the second part when it is supposed that $\theta_2(t) \in \mathcal{E}$, the result $\theta_2^{(n)}(t) \in \mathcal{E}$ $\forall n \in \mathbb{N}$ can immediately be obtained. Substituting this result into (13) yields $\theta_1(t) \in \mathcal{E}$, and hence, $\theta_1^{(n)}(t) \in \mathcal{E}$ $\forall n \in \mathbb{N}$. □

The propagation of vibration suppression principle suggests that when the trolley acceleration approaches zero, the suppression of the first swing angle implies that of the second swing angle. For this reason, information of the second swing angle can be excluded from the control law.

4. Augmented passivity-based control law design

An augmented passivity-based controller will be designed based on the propagation of vibration suppression principle. More specifically, the second swing angle is removed from the control law. A passivity analysis, which is presented in section 4.1, forms the basis for the proposed controller's structure (see section 4.2). The stability of the closed-loop system is then demonstrated.

4.1. Passivity analysis

It will be shown that the mapping from the trolley velocity $\dot{x}$ to the first swing angle θ_1 (denoted as $\dot{x} \mapsto \theta_1$) is lossless passive.

By applying (14) on the DPOC's linearized model (2), the decoupling dynamics of the first swing angle θ_1 can be obtained as follows

$$\begin{aligned} &m_1 l_1 l_2\theta_1^{(4)} + (m_1 + m_2)\,(l_1 + l_2)\, g\ddot{\theta}_1 + (m_1 + m_2)\, g^2\theta_1 \\ &\quad = m_1 l_2 x^{(4)} + (m_1 + m_2)\, g\ddot{x}. \end{aligned} \tag{15}$$

Denoting $\dot{\mathcal{X}}(s)$ and $\Theta_1(s)$ as Laplace transforms of the trolley velocity $\dot{x}(t)$ and the first swing angle $\theta_1(t)$, respectively. From (15), the transfer function $G_1(s)$ from $\dot{\mathcal{X}}(s)$ to $\Theta_1(s)$ can be computed as

$$G_1(s) = \frac{s\left[m_1 l_2 s^2 + (m_1 + m_2)\, g\right]}{m_1 l_1 l_2 s^4 + (m_1 + m_2)\,(l_1 + l_2)\, gs^2 + (m_1 + m_2)\, g^2}.$$

The state-space realization of $G_1(s)$ can be obtained by defining a state ξ_1, which satisfies the following scalar differential equation

$$m_1 l_1 l_2\xi_1^{(4)} + (m_1 + m_2)\,(l_1 + l_2)\, g\ddot{\xi}_1 + (m_1 + m_2)\, g^2\xi_1 = \dot{x}.$$

Moreover, let $\xi_2 := \dot{\xi}_1$, $\xi_3 := \dot{\xi}_2$, and $\xi_4 := \dot{\xi}_3$, and $\boldsymbol{\xi} := \begin{bmatrix}\xi_1 & \xi_2 & \xi_3 & \xi_4\end{bmatrix}^\top \in \mathbb{R}^4$, then

$$\begin{aligned}\begin{bmatrix}\dot{\xi}_1\\ \dot{\xi}_2\\ \dot{\xi}_3\\ \dot{\xi}_4\end{bmatrix} &= \begin{bmatrix}0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\\ \beta_1 & 0 & \beta_2 & 0\end{bmatrix}\begin{bmatrix}\xi_1\\ \xi_2\\ \xi_3\\ \xi_4\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 0\\ \beta_3\end{bmatrix}\dot{x}\\ \theta_1 &= \begin{bmatrix}0 & \beta_4 & 0 & \beta_5\end{bmatrix}\boldsymbol{\xi},\end{aligned} \tag{16}$$

where it is denoted that $\beta_1 := -(m_1+m_2)\,g^2/(m_1 l_1 l_2)$, $\beta_2 := -(m_1+m_2)(l_1+l_2)\,g/(m_1 l_1 l_2)$, $\beta_3 := 1/(m_1 l_1 l_2)$, $\beta_4 := (m_1+m_2)\,g$, and $\beta_5 := m_1 l_2$. Moreover, in (16), the control input is the trolley velocity $\dot{x}$ and the output is the first swing angle θ_1.

Theorem 3. *Consider the decoupling dynamics (15) with the state-space representation (16). The input–output mapping $\dot{x} \mapsto \theta_1$ is lossless passive.*

Proof. It is straightforward to verify that the storage function $S = 0.5\boldsymbol{\xi}^\top \boldsymbol{P}\boldsymbol{\xi}$ verifies the relation $\dot{S} = \dot{x}\theta_1$, in which $\boldsymbol{P} \in \mathbb{R}^{4\times 4}$ is a positive definite matrix of the form

$$\boldsymbol{P} = \begin{bmatrix}p_{11} & p_{12} & p_{13} & p_{14}\\ p_{12} & p_{22} & p_{23} & p_{24}\\ p_{13} & p_{23} & p_{33} & p_{34}\\ p_{14} & p_{24} & p_{34} & p_{44}\end{bmatrix}, \tag{17}$$

whose elements are given by

$$\begin{aligned}&p_{11} = -\beta_1 p_{24},\ p_{12} = 0,\ p_{13} = -\beta_1 p_{44},\ p_{14} = 0,\\ &p_{22} = -p_{13} - \beta_2 p_{24},\ p_{23} = 0,\ p_{24} = \beta_4/\beta_3,\\ &p_{33} = -p_{24} - \beta_2 p_{44},\ p_{34} = 0,\ p_{44} = \beta_5/\beta_3.\end{aligned} \tag{18}$$

This therefore completes the proof. □

Given that the mapping from the trolley velocity $\dot{x}$ to the first swing angle θ_1 is (lossless) passive, it is of interest to ask whether the same property holds for the mapping from trolley velocity $\dot{x}$ to the second swing angle θ_2 (denoted as $\dot{x} \mapsto \theta_2$). The answer is negative, and it can be explained as follows. By substituting (13) into (2), the decoupling dynamics of the second swing angle θ_2 is

$$\begin{aligned}&m_1 l_1 l_2 \theta_2^{(4)} + (m_1+m_2)(l_1+l_2)\,g\ddot{\theta}_2 + (m_1+m_2)\,g^2\theta_2\\ &\quad = (m_1+m_2)\,g\ddot{x}.\end{aligned} \tag{19}$$

Theorem 4. *Consider the system (19). The input–output mapping $\dot{x} \mapsto \theta_2$ is non-passive.*

Proof. Let $\Theta_2(s)$ be the Laplace transform of the second swing angle $\theta_2(t)$. The transfer function $G_2(s)$ from $\dot{\mathcal{X}}(s)$ to $\Theta_2(s)$ is given as follows

$$G_2(s) = \frac{(m_1+m_2)\,gs}{m_1 l_1 l_2 s^4 + (m_1+m_2)(l_1+l_2)\,gs^2 + (m_1+m_2)\,g^2}.$$

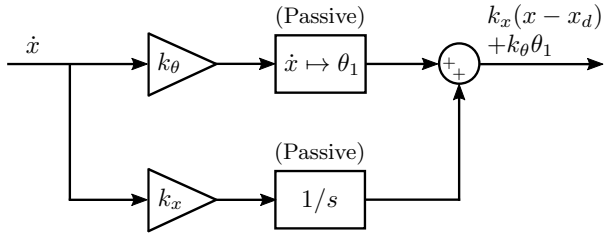

Figure 3: Passive parallel connection of the original mapping $\dot{x} \mapsto \theta_1$ and an integrator.

It can be seen that the relative degree of $G_2(s)$ is three, which is greater than one. As a result, $G_2(s)$ is not a positive real transfer function (Khalil, 2002, p. 238). Since for linear systems, positive realness is equivalent to passivity (Brogliato et al., 2020, p. 86), the mapping $\dot{x} \mapsto \theta_2$ is therefore non-passive. □

So far, it was shown that the mapping $\dot{x} \mapsto \theta_1$ is passive with the storage function $S = 0.5\boldsymbol{\xi}^\top \boldsymbol{P}\boldsymbol{\xi}$ (i.e., $\dot{S} = \dot{x}\theta_1$), where $\boldsymbol{P}$ is given by (17) and (18). It can be seen that the passive mapping $\dot{x} \mapsto \theta_1$ only contains the first swing angle information in its output. Therefore, if the passivity-based controller is simply designed as $\dot{x} = -k_\theta\theta_1$ ($k_\theta > 0$) to render $\dot{S} = -k_\theta\theta_1^2 \le 0$ (Khalil, 2002, p. 604), the control objective (3) cannot be achieved, because the trolley information is missing in the control law. In order to circumvent this problem, one can add a (passive) integrator parallel to the original mapping $\dot{x} \mapsto \theta_1$ as shown in Fig. 3. Since a parallel connection of two passive systems are passive (Haddad and Chellaboina, 2008, p. 458), the mapping $\dot{x} \mapsto k_\theta\theta_1 + k_x(x-x_d)$ is passive for $k_\theta > 0$ and $k_x > 0$. In fact, its storage function is given by

$$S_1 = k_\theta S + 0.5k_x(x-x_d)^2, \tag{20}$$

that is to say, $\dot{S}_1 = \dot{x}\left[k_\theta\theta_1 + k_x(x-x_d)\right]$. This result gives the idea for the structure of the augmented passivity-based control law, which will be designed in the next subsection.

4.2. Augmented passivity-based control law design and stability analysis

Since the mapping $\dot{x} \mapsto k_\theta\theta_1 + k_x(x-x_d)$ is passive, it is straightforward to choose the following (preliminary) passivity-based controller

$$\dot{x} = -k_\theta\theta_1 - k_x(x-x_d) \tag{21}$$

to render $\dot{S}_1 = -\left[k_\theta\theta_1 + k_x(x-x_d)\right]^2 \le 0$. However, the control law (21) might generate a discontinuous velocity command at the beginning when it is activated due to non-zero initial conditions. To address this problem, we propose to augment the control law (21) by an exponentially decaying function $f \in \mathcal{E}$ to compensate for the aforementioned initial mismatch in the trolley's velocity. Namely, the control law (21) is revised as follows

$$\begin{aligned}\dot{x} &= -k_\theta\theta_1 - k_x(x-x_d) + f\\ \dot{f} &= -\alpha f,\end{aligned} \tag{22}$$

where $\alpha > 0$ and the initial condition of the function f is chosen as $f(0) = \dot{x}(0) + k_x\,(x(0) - x_d) + k_\theta \theta_1(0)$.

Theorem 5. *Consider the decoupling dynamics (15) of the first swing angle θ_1 controlled by the augmented passivity-based control law (22). If $k_x > 0$, $k_\theta > 0$, and $\alpha > 0$, it is true that θ_1, $\dot{\theta}_1$, $x - x_d$, $\dot{x}$, and $\ddot{x}$ belong to $\mathcal{E}$.*

Proof. Denoting $\xi_5 := x - x_d$ and $\xi_6 := f$. By substituting the controller (22) into the state-space representation (16), it can be obtained that

$$\begin{bmatrix} \dot{\xi}_1 \\ \dot{\xi}_2 \\ \dot{\xi}_3 \\ \dot{\xi}_4 \\ \dot{\xi}_5 \\ \dot{\xi}_6 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ \lambda_1 & \lambda_2 & \lambda_3 & \lambda_4 & \lambda_5 & \lambda_6 \\ 0 & \lambda_7 & 0 & \lambda_8 & -k_x & 1 \\ 0 & 0 & 0 & 0 & 0 & -\alpha \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \\ \xi_4 \\ \xi_5 \\ \xi_6 \end{bmatrix}, \tag{23}$$

where it is denoted that $\lambda_1 := -(m_1 + m_2)\,g^2/(m_1 l_1 l_2)$, $\lambda_2 := -k_\theta\,(m_1 + m_2)\,g/(m_1 l_1 l_2)$, $\lambda_4 := -k_\theta/l_1$, $\lambda_5 := -k_x/(m_1 l_1 l_2)$, $\lambda_3 := -(m_1 + m_2)\,(l_1 + l_2)\,g/(m_1 l_1 l_2)$, $\lambda_6 := 1/(m_1 l_1 l_2)$, $\lambda_7 := -k_\theta\,(m_1 + m_2)\,g$, and $\lambda_8 := -k_\theta m_1 l_2$. By using the Hurwitz criterion, it is straightforward to check that all roots of the characteristic polynomial associating with (23) have negative real parts when $k_x > 0$, $k_\theta > 0$, and $\alpha > 0$. Thus, the closed-loop system (23) is globally asymptotically stable. As a result, $\xi_i \in \mathcal{E}$ $(i = 1, \ldots, 6)$ (Khalil, 2002, p. 134). Since $\xi_5 = x - x_d$ by definition, $x - x_d \in \mathcal{E}$. From (16), $\theta_1 = (m_1 + m_2)\,g\xi_2 + m_1 l_2 \xi_4$, and thus, $\theta_1 \in \mathcal{E}$. Because $\theta_1, x - x_d \in \mathcal{E}$, it is true that $\dot{\theta}_1, \dot{x}, \ddot{x} \in \mathcal{E}$. This therefore completes the proof. □

The following theorem demonstrates that all control objectives and requirements described at the end of section 2 can be realized by the controller (22).

Theorem 6. *The nonlinear DPOC system (1) controlled by the augmented passivity-based control law (22) is locally asymptotically stable. In particular, locally, it is true that $x - x_d$, $\dot{x}$, θ_1, $\dot{\theta}_1$, θ_2, and $\dot{\theta}_2$ belong to $\mathcal{L}_\infty^{\to 0}$.*

Proof. The desired result can be proved using the propagation of vibration suppression principle and the Lyapunov linearization method. Recalling that the linearized dynamics of the nonlinear model (1) is given in (2). Differentiating the control law (22) once and substituting the result into (2), the resulting closed-loop system is fully autonomous, whose states are $x - x_d$, $\dot{x}$, θ_i, $\dot{\theta}_i$ $(i = 1, 2)$, f, and $\dot{f}$. It was shown in Theorem 5 that $x - x_d$, $\dot{x}$, $\ddot{x}$, θ_1, $\dot{\theta}_1$, f, $\dot{f} \in \mathcal{E}$. Hence, we have $\theta_1, \ddot{x} \in \mathcal{E}$, implying that $\theta_2, \dot{\theta}_2 \in \mathcal{E}$ due to statement 2) of the propagation of vibration suppression principle described in Theorem 2. Thus, the linearized closed-loop system is globally asymptotically stable. This indicates that the nonlinear closed-loop system (governed by the nonlinear DPOC dynamics (1) and the augmented passivity-based control law (22)) is locally asymptotically stable, as a result of the Lyapunov linearization theorem (Khalil, 2002, p. 139). For this reason, locally, the states $x - x_d$, $\dot{x}$, θ_1, $\dot{\theta}_1$, θ_2, and $\dot{\theta}_2$ belong to $\mathcal{L}_\infty^{\to 0}$. □

Remark 1. It is indeed possible to use the second swing angle θ_2 instead of the first swing angle θ_1 in the stabilizing control law. However, the first swing angle is chosen due to the following reasons:

- The input–output mapping $\dot{x} \mapsto \theta_1$ has a nice passive property, which makes the controller design and analysis easy. In contrast, the mapping $\dot{x} \mapsto \theta_2$ does not have such a characteristic.
- It is easier to measure the first swing angle θ_1 than the second swing angle θ_2 in practice.

4.3. Computation of ROA

In Theorem 6, only local stability has qualitatively been shown. In this section, it is of interest to estimate the ROA for the closed-loop nonlinear model. Except for low-order systems, one often needs to resort numerical methods to compute the ROA (Chesi, 2004). Inspired from the work in (Chesi, 2009), the SOS (convex) optimization framework will be employed to compute the ROA. Denoting $\delta := \theta_1 - \theta_2$ as the relative swing angle. It is found that, by using δ instead of θ_2, a less conservative estimate of ROA can be obtained. Let $\boldsymbol{\Theta} := [\theta_1\ \delta]^\top \in \mathbb{R}^2$ and $\boldsymbol{q} := [\dot{x}\ \dot{f}\ \boldsymbol{\Theta}^\top\ \dot{\boldsymbol{\Theta}}^\top]^\top \in \mathbb{R}^6$ be vectors of swing angles and system states, respectively. The original nonlinear system (1) controlled by the control law (22) is explicitly given by

$$\begin{aligned} &\boldsymbol{M}(\boldsymbol{\Theta})\ddot{\boldsymbol{\Theta}} + \boldsymbol{C}(\boldsymbol{\Theta}, \dot{\boldsymbol{\Theta}}) + \boldsymbol{G}(\boldsymbol{\Theta}) = \boldsymbol{H}(\boldsymbol{\Theta})\ddot{x} \\ &\dot{x} = -k_x(x - x_d) - k_\theta \theta_1 + f, \end{aligned} \tag{24}$$

in which the matrices are

$$\boldsymbol{M}(\boldsymbol{\Theta}) = \begin{bmatrix} (m_1 + m_2)l_1 + m_2 l_2 \cos\delta & -m_2 l_2 \cos\delta \\ l_1 \cos\delta + l_2 & -l_2 \end{bmatrix}$$

$$\boldsymbol{C}(\boldsymbol{\Theta}, \dot{\boldsymbol{\Theta}}) = \begin{bmatrix} m_2 l_2 (\dot{\theta}_1 - \dot{\delta})^2 \sin\delta \\ -l_1 \dot{\theta}_1^2 \sin\delta \end{bmatrix}$$

$$\boldsymbol{G}(\boldsymbol{\Theta}) = \begin{bmatrix} (m_1 + m_2) g \sin\theta_1 \\ g \sin(\theta_1 - \delta) \end{bmatrix}, \ \boldsymbol{H}(\boldsymbol{\Theta}) = \begin{bmatrix} (m_1 + m_2)\cos\theta_1 \\ \cos(\theta_1 - \delta) \end{bmatrix}.$$

Let $\boldsymbol{d_0} = [1\ 0]^\top$, the linearized dynamics of (24) around the origin $\boldsymbol{q} = \boldsymbol{0}_{6\times 1}$ is given as follows

$$\dot{\boldsymbol{q}} = \boldsymbol{A}\boldsymbol{q}, \text{ where} \tag{25}$$

$$\boldsymbol{A} = \begin{bmatrix} -k_x & 1 & \boldsymbol{0}_{1\times 2} & -k_\theta \boldsymbol{d_0}^\top \\ 0 & -\alpha & \boldsymbol{0}_{1\times 2} & \boldsymbol{0}_{1\times 2} \\ \boldsymbol{0}_{2\times 1} & \boldsymbol{0}_{2\times 1} & \boldsymbol{0}_{2\times 2} & \boldsymbol{I}_{2\times 2} \\ -k_x \widetilde{\boldsymbol{M}}^{-1}\widetilde{\boldsymbol{H}} & \widetilde{\boldsymbol{M}}^{-1}\widetilde{\boldsymbol{H}} & -\widetilde{\boldsymbol{M}}^{-1}\widetilde{\boldsymbol{G}}_{\boldsymbol{0}} & -k_\theta \widetilde{\boldsymbol{M}}^{-1}\widetilde{\boldsymbol{H}}\boldsymbol{d_0}^\top \end{bmatrix}$$

$$\widetilde{\boldsymbol{M}} = \begin{bmatrix} (m_1 + m_2)l_1 + m_2 l_2 & -m_2 l_2 \\ l_1 + l_2 & -l_2 \end{bmatrix}$$

$$\widetilde{\boldsymbol{G}}_{\boldsymbol{0}} = \begin{bmatrix} (m_1 + m_2)g & 0 \\ g & -g \end{bmatrix}, \ \widetilde{\boldsymbol{H}} = \begin{bmatrix} m_1 + m_2 \\ 1 \end{bmatrix}.$$

From the proof of Theorem 6, it is known that the linearized system (25) is stable, and hence, for a chosen positive definite matrix $\boldsymbol{Q} \in \mathbb{R}^{6\times 6}$, there exists a positive definite matrix $\boldsymbol{\Phi} \in \mathbb{R}^{6\times 6}$ such that $\boldsymbol{\Phi}\boldsymbol{A} + \boldsymbol{A}^\top\boldsymbol{\Phi} = -\boldsymbol{Q}$ (Khalil, 2002, p. 136). Let $\eta := l_1 l_2(m_1 + m_2 \sin^2\delta) > 0$. Choosing $V(\boldsymbol{q}) = \boldsymbol{q}^\top\boldsymbol{\Phi}\boldsymbol{q}$ as a Lyapunov function for (24), then

$$\eta\dot{V} = \left(\frac{\partial V}{\partial \boldsymbol{q}}\right)^\top \Bigg(\boldsymbol{F}_1(m_1 + 0.5m_2(1-\cos 2\delta)) + \begin{bmatrix}\boldsymbol{0}_{4\times 1}\\ -\boldsymbol{M_0 C}\end{bmatrix} + \begin{bmatrix}\boldsymbol{0}_{4\times 1}\\ -\boldsymbol{M_0 G}\end{bmatrix} + \begin{bmatrix}\boldsymbol{0}_{4\times 1}\\ \boldsymbol{M_0 H}\end{bmatrix}\ddot{x}\Bigg), \quad (26)$$

in which $\boldsymbol{F}_1 := l_1 l_2[\ddot{x} \;\; -\alpha\dot{f} \;\; \dot{\boldsymbol{\Theta}}^\top \;\; \boldsymbol{0}_{1\times 2}]^\top \in \mathbb{R}^6$, $\boldsymbol{M_0} := \eta\boldsymbol{M}^{-1} \in \mathbb{R}^{2\times 2}$, and $\ddot{x} = -k_x\dot{x} - k_\theta\dot{\theta}_1 + \dot{f}$. By denoting $\boldsymbol{F_{C}}_j := [\boldsymbol{0}_{1\times 4} \;\; \boldsymbol{f_{C}}_j]^\top \in \mathbb{R}^6$ $(j = 1, 2)$, $\boldsymbol{F_{G}}_i := [\boldsymbol{0}_{1\times 4} \;\; \boldsymbol{f_{G}}_i]^\top \in \mathbb{R}^6$, and $\boldsymbol{F_{H}}_i := [\boldsymbol{0}_{1\times 4} \;\; \boldsymbol{f_{H}}_i]^\top \in \mathbb{R}^6$ $(i = 1, 2, 3)$, where

$$\begin{aligned}
&\boldsymbol{f_{C}}_1 = -\left[m_2 l_2^2(\dot{\theta}_1 - \dot{\delta})^2 \quad m_2 l_2^2(\dot{\theta}_1 - \dot{\delta})^2 + (m_1 + m_2) l_1^2\dot{\theta}_1^2\right]\\
&\boldsymbol{f_{C}}_2 = -0.5m_2 l_1 l_2\left[\dot{\theta}_1^2 \quad (\dot{\theta}_1 - \dot{\delta})^2 + \dot{\theta}_1^2\right]\\
&\boldsymbol{f_{H}}_1 = (m_1 + 0.5m_2) l_2\left[1 \quad 1\right]\\
&\boldsymbol{f_{H}}_2 = 0.5(m_1 + m_2) l_1\left[0 \quad 1\right],\ \boldsymbol{f_{H}}_3 = -0.5m_2 l_2\left[1 \quad 1\right]\\
&\boldsymbol{f_{G}}_1 = -g\boldsymbol{f_{H}}_1,\ \boldsymbol{f_{G}}_2 = g\boldsymbol{f_{H}}_2,\ \boldsymbol{f_{G}}_3 = -g\boldsymbol{f_{H}}_3,
\end{aligned}$$

the following results hold true

$$\begin{aligned}
\begin{bmatrix}\boldsymbol{0}_{4\times 1}\\ -\boldsymbol{M_0 C}\end{bmatrix} &= \boldsymbol{F_{C}}_1 \sin\delta + \boldsymbol{F_{C}}_2 \sin 2\delta\\
\begin{bmatrix}\boldsymbol{0}_{4\times 1}\\ -\boldsymbol{M_0 G}\end{bmatrix} &= \boldsymbol{F_{G}}_1 \sin\theta_1 + \boldsymbol{F_{G}}_2 \sin(\theta_1 - \delta)\\
&\quad -\boldsymbol{F_{G}}_2 \sin(\theta_1 + \delta) + \boldsymbol{F_{G}}_3 \sin(\theta_1 - 2\delta)\\
\begin{bmatrix}\boldsymbol{0}_{4\times 1}\\ \boldsymbol{M_0 H}\end{bmatrix} &= \boldsymbol{F_{H}}_1 \cos\theta_1 + \boldsymbol{F_{H}}_2 \cos(\theta_1 + \delta)\\
&\quad -\boldsymbol{F_{H}}_2 \cos(\theta_1 - \delta) + \boldsymbol{F_{H}}_3 \cos(\theta_1 - 2\delta).
\end{aligned}$$

Defining $\underline{(.)} := \min\{0, (.)\}$ and $\overline{(.)} := \max\{0, (.)\}$. By means of the Taylor theorem (Sherbert and Bartle, 2011, p. 189), for a function $y : \mathbb{R} \mapsto \mathbb{R}$, we have

$$\begin{aligned}
\sin y &= y - (1/6)\cos(\varepsilon_y) y^3\\
\cos y &= 1 - (1/2)\cos(\varepsilon_y) y^2,
\end{aligned}$$

where $\varepsilon_y \in (\underline{y},\ \overline{y})$. By applying these Taylor expansions on (26), the following result holds

$$\eta\dot{V} = \underbrace{\left(\frac{\partial V}{\partial \boldsymbol{q}}\right)^\top \boldsymbol{p}(\boldsymbol{q})}_{\chi(\boldsymbol{q})} + \underbrace{\left[\left(\frac{\partial V}{\partial \boldsymbol{q}}\right)^\top \boldsymbol{h}_1 \quad \ldots \quad \left(\frac{\partial V}{\partial \boldsymbol{q}}\right)^\top \boldsymbol{h}_{11}\right]}_{\boldsymbol{h}^\top(\boldsymbol{q})} \boldsymbol{w}, \quad (27)$$

where $\boldsymbol{w} = [w_1 \;\; \ldots \;\; w_{11}]^\top \in \mathbb{R}^{11}$ and $w_i := \cos\varepsilon_i$ $(i = 1, \ldots, 11)$ with $\varepsilon_1 \in (\underline{2\delta},\ \overline{2\delta})$, $\varepsilon_2 \in (\underline{\delta},\ \overline{\delta})$, $\varepsilon_3 \in (\underline{2\delta},\ \overline{2\delta})$, $\varepsilon_4 \in (\underline{\theta_1},\ \overline{\theta_1})$, $\varepsilon_5 \in (\underline{\theta_1 - \delta},\ \overline{\theta_1 - \delta})$, $\varepsilon_6 \in (\underline{\theta_1 + \delta},\ \overline{\theta_1 + \delta})$, $\varepsilon_7 \in (\underline{\theta_1 - 2\delta},\ \overline{\theta_1 - 2\delta})$, $\varepsilon_8 \in (\underline{\theta_1},\ \overline{\theta_1})$, $\varepsilon_9 \in (\underline{\theta_1 + \delta},\ \overline{\theta_1 + \delta})$, $\varepsilon_{10} \in (\underline{\theta_1 - \delta},\ \overline{\theta_1 - \delta})$, and $\varepsilon_{11} \in (\underline{\theta_1 - 2\delta},\ \overline{\theta_1 - 2\delta})$. Moreover, the remaining terms in (27) are defined by

$$\begin{aligned}
\boldsymbol{p}(\boldsymbol{q}) = {} & \boldsymbol{F}_1 m_1 + [\boldsymbol{F_{C}}_1 + 2\boldsymbol{F_{C}}_2 - 2(\boldsymbol{F_{G}}_2 + \boldsymbol{F_{G}}_3)]\,\delta\\
& + (\boldsymbol{F_{G}}_1 + \boldsymbol{F_{G}}_3)\theta_1 + (\boldsymbol{F_{H}}_1 + \boldsymbol{F_{H}}_3)\ddot{x},\\
\boldsymbol{h}_1 = {} & \boldsymbol{F}_1 m_2\delta^2,\ \boldsymbol{h_2} = (-1/6)\boldsymbol{F_{C}}_1\delta^3,\ \boldsymbol{h}_3 = (-4/3)\boldsymbol{F_{C}}_2\delta^3,\\
\boldsymbol{h}_4 = {} & (-1/6)\boldsymbol{F_{G}}_1\theta_1^3,\ \boldsymbol{h}_5 = (-1/6)\boldsymbol{F_{G}}_2(\theta_1 - \delta)^3,\\
\boldsymbol{h}_6 = {} & (1/6)\boldsymbol{F_{G}}_2(\theta_1 + \delta)^3,\ \boldsymbol{h}_7 = (-1/6)\boldsymbol{F_{G}}_3(\theta_1 - 2\delta)^3,\\
\boldsymbol{h}_8 = {} & (-1/2)\boldsymbol{F_{H}}_1\theta_1^2\ddot{x},\ \boldsymbol{h}_9 = (-1/2)\boldsymbol{F_{H}}_2(\theta_1 + \delta)^2\ddot{x},\\
\boldsymbol{h}_{10} = {} & (1/2)\boldsymbol{F_{H}}_2(\theta_1 - \delta)^2\ddot{x},\ \boldsymbol{h}_{11} = (-1/2)\boldsymbol{F_{H}}_3(\theta_1 - 2\delta)^2\ddot{x}.
\end{aligned}$$

Let $\mathcal{C} = \left\{\boldsymbol{q} \in \mathbb{R}^6 : \theta_1^2 + \delta^2 \le r_{\max}^2\right\}$, where $r_{\max} > 0$ is a chosen parameter. Furthermore, for a given constant $c > 0$, denoting $\Omega_c = \left\{\boldsymbol{q} \in \mathbb{R}^6 : V(\boldsymbol{q}) \le c\right\}$ as the sublevel set of the Lyapunov function $V(\boldsymbol{q})$. Defining $\Gamma_c = \Omega_c \cap \mathcal{C}$. Then, Γ_c is the ROA of the nonlinear closed-loop system (24) if the condition $\boldsymbol{q} \in \Gamma_c \implies \dot{V}(\boldsymbol{q}) \le 0$ holds, as a result of Lasalle invariance theorem (Khalil, 2002, p. 128). Such a condition can be encoded in a SOS optimization program by finding a scalar polynomial s_p such that

$$s_p > 0 \text{ and } -\left(\chi(\boldsymbol{q}) + \boldsymbol{h}^\top(\boldsymbol{q})\widetilde{\boldsymbol{w}}\right) + (V(\boldsymbol{q}) - c)\, s_p > 0, \quad (28)$$

for all $\widetilde{\boldsymbol{w}} \in \mathrm{ver}(W)$, in which the inequalities should be understood in the sense of SOS. Moreover, $\mathrm{ver}(W)$ is the set of vertices of the 11-dimensional rectangle $W = [\sigma_{1-} \;\; \sigma_{1+}] \times \ldots \times [\sigma_{11-} \;\; \sigma_{11+}]$, where σ_{i-} and σ_{i+} $(i = 1, \ldots, 11)$ are the lower and upper bounds of w_i over the region defined by the set $\mathcal{C}$.

By choosing a positive definite matrix $\boldsymbol{Q}$, the positive definite matrix $\boldsymbol{\Phi}$ can be determined by the Lyapunov equation $\boldsymbol{\Phi}\boldsymbol{A} + \boldsymbol{A}^\top\boldsymbol{\Phi} = -\boldsymbol{Q}$. By further selecting $r_{\max} > 0$ and $c > 0$, the bounds σ_{i-} and σ_{i+} can be computed, and then, the SOS feasibility test (28) can be conducted at $2^{11} = 2048$ vertices of W. If the result is positive, Γ_c is the ROA. Otherwise, the value of c can be adjusted using the bisection method. However, to optimally determine the ROA, one can perform a diagonal matrix parameterization $\boldsymbol{Q} = \mathrm{diag}(\mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6) \in \mathbb{R}^{6\times 6}$, where $\mu_i > 0$ $(i = 1, \ldots, 6)$ are to-be-optimized variables. The following optimization problem is then configured

$$\begin{aligned}
\max_{\mu_i > 0,\, c > 0} \quad & c/\lambda_{\max}(\boldsymbol{\Phi}) \qquad (29)\\
\text{s.t.} \quad & \text{The SOS program (28) is feasible,}
\end{aligned}$$

where $\lambda_{\max}(\boldsymbol{\Phi}) > 0$ is the maximum eigenvalue of $\boldsymbol{\Phi}$. As an example, by using the system parameters shown in section 5 (i.e., $m_1 = 2$ kg, $m_2 = 5$ kg, $l_1 = 1$ m, and $l_2 = 0.5$ m), the particle swarm optimization routine (Tho and Terashima, 2019) can be employed to solve (29), in which the SOS feasibility test (28) can be carried out by using the SOSTOOLS parser and the MOSEK solver. It

Table 1: Optimization results.

	Optimal cost $J^\star$	Optimal gains
Proposed controller (22)	1.93	$k_x = 0.65$, $k_\theta = 6.16$, $\alpha = 1.06$.
Sun et al., 2019	2.02	$k_p = 0.56$, $k_d = 1.13$, $\mu = 0.47$.
Tuan and Lee, 2013	1.89	$k_1 = 4.31$, $k_2 = 2.33$, $k_3 = 48.59$, $k_4 = -12.32$, $K_a = 9.83$.
Yang et al., 2022	1.91	$\nu_1 = 0.54$, $\nu_2 = 8.92$, $\nu_3 = 9.01$, $K_s = 0.15$, $\alpha_{1,2} = 1.02$, $\alpha_{1,3} = -0.21$, $\Lambda = 7.48$, $\kappa = 0.35$.

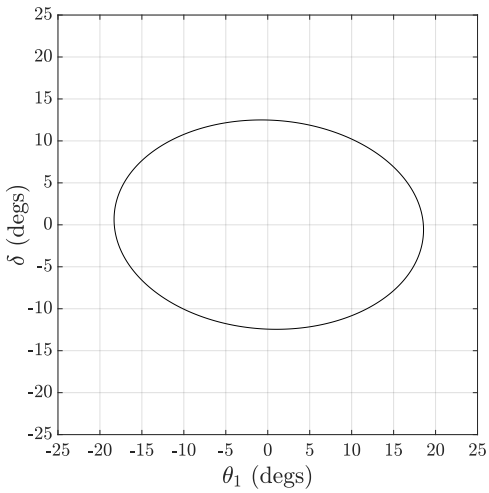

Figure 4: Visualization of the ROA Γ_c when $x(0) = \dot{x}(0) = 0$, $\dot{\theta}_1(0) = \dot{\delta}(0) = 0.2$ rad/s (or 11.5 degs/s), and $x_d = 1$ m.

is also selected that $r_{\max} = 0.35$ rad (or 20 degs). The optimal results are $\mu_1^\star = 1.36$, $\mu_2^\star = 1.07 \times 10^{-5}$, $\mu_3^\star = 125.5$, $\mu_4^\star = 133.67$, $\mu_5^\star = 8.98$, $\mu_6^\star = 1.47$, and $c^\star = 8.02$. Since the ROA Γ_c is in 6 dimensions, it is not possible to visualize it. However, Γ_c can be visualized on a planar plane when its four out of fix dimensions are fixed. For instance, the ROA Γ_c is demonstrated in Fig. 4 when $x(0) = \dot{x}(0) = 0$, $\dot{\theta}_1(0) = \dot{\delta}(0) = 0.2$ rad/s (or 11.5 degs/s), and $x_d = 1$ m. It can be observed that the area of the ROA Γ_c is sufficiently large, and thus, it is practically acceptable.

Remark 2. An analogous result to the propagation of vibration suppression principle (as introduced in Theorem 2) can be deduced for double *inverted* pendulum systems. However, a controller that employs only the first or only the second pendulum angle to stabilize the double inverted pendulum system yields a very small region of attraction for the closed-loop dynamics, rendering it impractical. Therefore, unlike the DPOC case, the angle information of both pendulums should be utilized to effectively control the double inverted pendulum system.

5. Comparative simulation result

In this section, the augmented passivity-based control law (22) will be compared with the following full-state feedback controllers (which utilize both swing angles):

1. Full-state feedback passivity-based controller (Sun et al., 2019)

$$\begin{aligned}\ddot{x} = &- k_p\left[(x - x_d) + \mu\left(\beta \sin\theta_1 + \sin\theta_2\right)\right] \\ &- k_d\left[\dot{x} + \mu\left(\beta\dot{\theta}_1\cos\theta_1 + \dot{\theta}_2\cos\theta_2\right)\right],\end{aligned}$$

   where $k_p > 0$, $k_d > 0$, and $\mu > 0$ are control gains. Moreover, $\beta := (m_1 + m_2)\, l_1 / (m_2 l_2)$.
2. Full-state feedback sliding-mode controller (SMC) (Tuan and Lee, 2013)

$$\ddot{x} = k_1^{-1}\left[-k_2\dot{x} - k_3\dot{\theta}_1 - k_4\dot{\theta}_2 - K_a \mathrm{sat}(\sigma)\right],$$

   where the sliding function is $\sigma := k_1\dot{x} + k_2(x - x_d) + k_3\theta_1 + k_4\theta_2$. The control gains are $k_1, k_2, k_3, K_a > 0$ and $k_4 < 0$. In addition, to avoid the chattering effect, the saturation function is used instead of the signum function. Such a function is defined as $\mathrm{sat}(\sigma) := \sigma/\epsilon$ if $|\sigma/\epsilon| < 1$, $\mathrm{sat}(\sigma) := -1$ if $\sigma/\epsilon < -1$, and $\mathrm{sat}(\sigma) = 1$ if $\sigma/\epsilon > 1$, where $\epsilon = 0.05$ is chosen.
3. Full-state feedback adaptive fuzzy controller (Yang et al., 2022)

$$\ddot{x} = -K\phi - \widehat{\boldsymbol{\rho}}^\top \boldsymbol{\varepsilon}_{\boldsymbol{z}}(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{x})\mathrm{sat}(\phi) - \widehat{\omega}\mathrm{sat}(\phi) - K_s\mathrm{sat}(\phi),$$

   where $\phi = \dot{x} + \nu_1(x - x_d) + \alpha_{1,2}(\dot{\theta}_1 + \nu_2\theta_1) + \alpha_{1,3}(\dot{\theta}_2 + \nu_3\theta_2)$; $\boldsymbol{\varepsilon}_{\boldsymbol{z}}(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{x}) \in \mathbb{R}^b$ is the vector of fuzzy basis functions with $b \in \mathbb{N}^+$ being the number of fuzzy rules; $\dot{\widehat{\boldsymbol{\rho}}} = \mathrm{proj}(\boldsymbol{\Psi})$ where $\boldsymbol{\Psi} = \Lambda\boldsymbol{\varepsilon}_{\boldsymbol{z}}(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{x})\phi$ with proj is the projection operator; and $\dot{\widehat{\omega}} = \kappa\phi\mathrm{sat}(\phi)$. The control gains are $\nu_1 > 0$, $\nu_2 > 0$, $\nu_3 > 0$, $\alpha_{1,2} > 0$, $\alpha_{1,3} < 0$, $K_s > 0$, $\Lambda > 0$, and $\kappa > 0$. Note also that, the membership functions of $\boldsymbol{\varepsilon}_{\boldsymbol{z}}(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{x})$ are identical to those of Eq. (44) in Yang et al., 2022, and $b = 7$.

To draw a fair comparison, the control gains of all controllers are optimized based on the following cost function

$$J = \kappa_1 \int_0^{20} (x - x_d)^2\, d\tau + \kappa_2 \int_0^{20} \theta_1^2 d\tau + \kappa_3 \int_0^{20} \theta_2^2 d\tau,$$

in which $\kappa_1 = 1$ and $\kappa_2 = \kappa_3 = 100$. The system parameters are $m_1 = 2$ kg, $m_2 = 5$ kg, $l_1 = 1$ m, and $l_2 = 0.5$ m. The trolley's reference is chosen as $x_d = 1$ m. The system is assumed to be at rest at the beginning, i.e.,

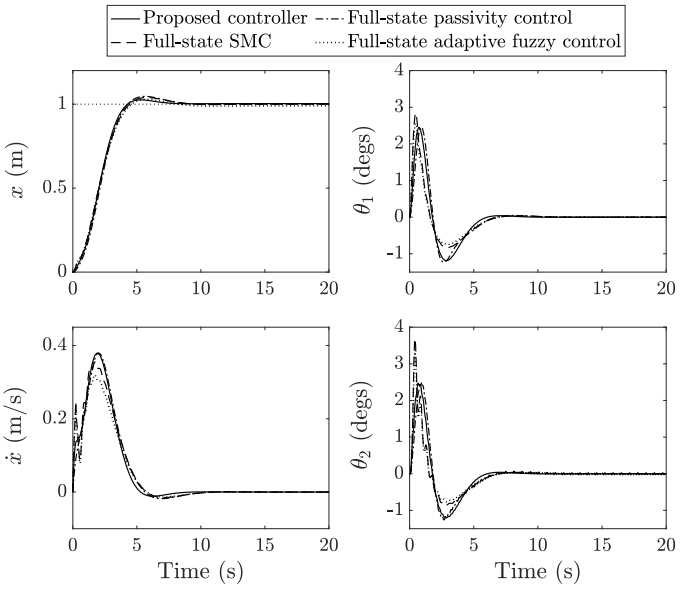

Figure 5: Comparative simulation result.

$x(0) = \dot{x}(0) = \theta_1(0) = \dot{\theta}_1(0) = \theta_2(0) = \dot{\theta}_2(0) = 0$. The particle swarm optimization mechanism is utilized to find optimal gains (Tho and Terashima, 2019).

The optimization results are aggregated in Table 1. The obtained optimal cost of the proposed controller (22) is $J^\star = 1.93$, which is relatively similar to that of the full-state feedback passivity-based controller (with $J^\star = 2.02$), the full-state feedback SMC (with $J^\star = 1.89$), and the full-state feedback adaptive fuzzy controller (with $J^\star = 1.91$). The comparative simulation result is shown in Fig. 5. Based on all of these evaluations, it can be concluded that performance of the proposed control law does not degrade in the absence of the second swing angle information when compared with existing full-state feedback controllers. This justified the main claim of the paper.

6. Experimental result

The augmented passivity-based controller (22) is now validated on a small-scale DPOC system, which is illustrated in Fig. 6. The actuator is a YASKAWA USAFED-09C2L1K AC servo motor and the servo driver is a MITSUBISHI MR-J2S-200A, which operates in velocity control mode. A universal serial bus (USB) control unit is utilized to convert analog signals to digital ones and vice versa. Matlab/Simulink is employed to build an executing file via Real Time Workshop. A Fedora Linux PC operates the executing file in real time with a sampling period of 10 ms via the real-time application interface. To measure the first swing angle, a pair of laser sensors KEYENCE VG-035 is used. A Window PC receives the second swing angle measurement from an inertial measurement unit (IMU) TSND121 of ATR Promotions via Bluetooth, and this information is sent to the Linux PC through user diagram

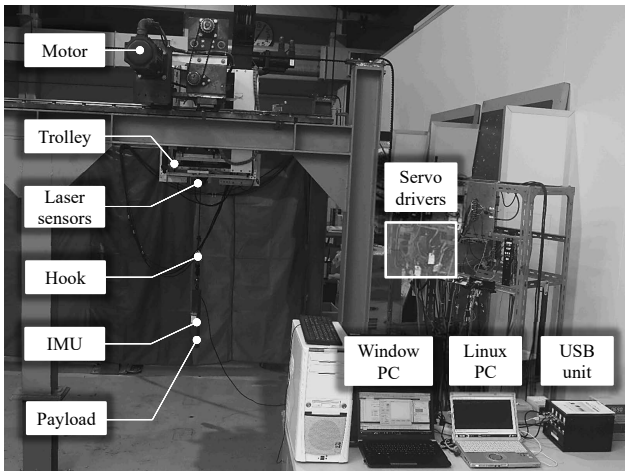

Figure 6: A small-scale DPOC rig.

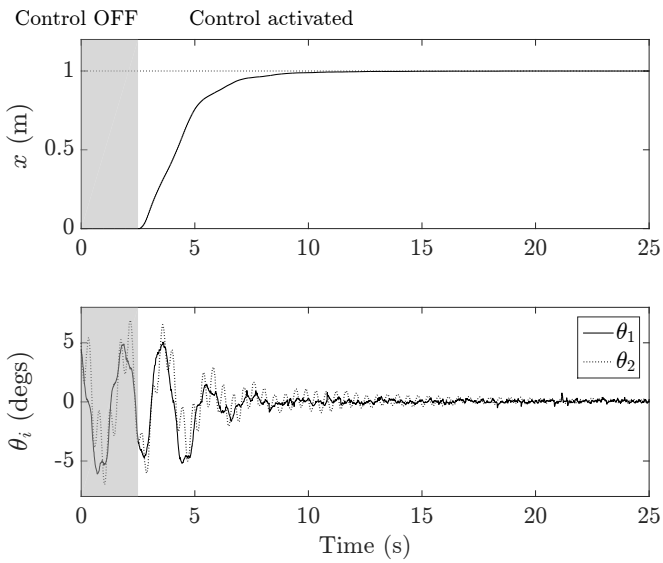

Figure 7: Experimental result of the augmented passivity-based controller.

protocol (UDP). Note that, the second swing angle measurement is only used for certifying the vibration suppression capability of the proposed controller, i.e., it is not employed in the control law.

To clearly demonstrate the efficacy of the established controller, an initial vibration is imposed. During the first 2.5 seconds, both pendulums freely vibrate and their amplitudes peak around 5 degrees. In the course of this free-motion (or uncontrolled period), the double-pendulum effect can be clearly observed from Fig. 7. At the time instant $t = 2.5$ seconds, the controller (22) is activated. It can be seen that the trolley precisely arrives at the destination while both swing angles are completely suppressed, confirming the effectiveness of the proposed approach.

It is also worth emphasizing that the implementations of the comparative full-state feedback controllers (as introduced in section 5) are not successful on our overhead crane rig. The reason is that the trolley velocity signal

$\dot{x}(t)$ obtained by differentiating the measured trolley position $x(t)$ is severely distorted on our (outdated) hardware platform. This signal distortion is not related to usual noises but the incompatibility between AD/DA converters of the (modern) USB unit and encoders of (legacy) servo motors. Thus, adding low-pass filters is not helpful in this case. Such an issue actually highlights the advantage of the proposed controller over the full-state feedback ones.

7. Discussion on future developments

There are several challenges that can be addressed in the future works:

1. Parametric uncertainties in system parameters (including the masses m_1, m_2 and the lengths l_1, l_2) were not considered in the computation of ROA as described in section 4.3. One way to resolve this challenge is to include the lower and upper bounds of each uncertain system parameter into the set of vertices ver(W), on which the SOS feasibility test is conducted. However, this method exponentially increases the problem size of the SOS program. This issue was not successfully resolved at present.
2. Even though the resulting estimate of ROA using an optimal quadratic Lyapunov function is practically acceptable, it is still somewhat conservative. This challenge can theoretically be solved by considering a higher order Lyapunov function, e.g., a quartic or sextic one. Nevertheless, the exponential growth in the computational complexity is, again, a major problem.
3. The stabilizing controller for the DPOC system, which excludes the second swing angle information, should be designed in a more systematic manner. This might open the door for controlling more complex underactuated and/or vibratory systems (e.g., N-pendulum overhead crane) without needing information of all unactuated coordinates as suggested by the propagation of vibration suppression principle.

8. Conclusion

An augmented passivity-based control law, which does not employ the second swing angle information, was established for a DPOC system. The basic concept leading to the elimination of the second swing angle measurement/estimation was the propagation of vibration suppression principle. It has been demonstrated that, despite the absence of the second swing angle, control performance of the proposed approach was comparable to that of existing full-state feedback controllers. Through both simulation and experimental results, efficacy of the developed scheme has been confirmed. Several possible future developments have also been discussed.

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Ho Duc Tho received the B.E. degree in mechatronics engineering (honors program) from the Ho Chi Minh City University of Technology, Ho Chi Minh City, Vietnam, in 2015, and the M.E. and D.Eng degrees in mechanical engineering from the Toyohashi University of Technology, Toyohashi, Japan, in 2017 and 2020, respectively.

From 2020 to 2025, he was with the Nagaoka University of Technology as an assistant professor. He is now with the Ho Chi Minh City University of Technology, Vietnam. Mr. Tho received the Japanese Government Monbukagakusho Scholarship (Top Global Project) in 2017 for his M.E. degree.

His current research interests include vibration control for flexible systems and telerobotics. He is a member of HMI Technical Committee (IEEE Consumer Technology Society).

Kazuhiko Terashima received the B.S. and M.S. degrees from the Kyoto Institute of Technology, Kyoto, Japan, in 1976 and 1978, respectively, and the Ph.D. degree in mechanical engineering from the Kyoto University, Kyoto, Japan, in 1982.

Since 1982, he had been with the Toyohashi University of Technology, Aichi, Japan, where he became a Professor in 1994 and served as President from 2020 to 2024. He was a Visiting Research Fellow with the Technical University of Munich, Munich, Germany, from September 1991 until September 1992. He passed away in May 2024.

During his lifetime, his research interests were in the analysis and design of control systems, with applications to motion control, robotics, and intelligent systems.

Takanori Miyoshi received the B.E. degree in electrical engineering from the Osaka University, Osaka, Japan, in 1989, and the M.E. and D.Eng degrees in mechanical engineering from the Toyohashi University of Technology, Toyohashi, Japan, in 1998 and 2001, respectively.

From 1989 to 2002, he worked at Roland Digital Group Corporation. In 2002, he joined the Toyohashi University of Technology as a lecturer, where he was later promoted to associate professor. Since 2019, he has been with the Nagaoka University of Technology as a full professor. Mr. Miyoshi was a visiting scholar at the Technical University of Munich, Munich, Germany, from 2005 to 2006.

His current research interests include telerobotics, power assistive system, vibration control, and system safety. He is a member of IEEE, JSME, RSJ, and SICE.